Wind Gusts and Rural Traffic Accidents in Iceland

A Reproducible Frequency-Adjusted Analysis, 2007–2024

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Abstract

This thesis examines whether strong wind gusts are disproportionately common when rural injury accidents occur in Iceland. A reproducible Python pipeline integrates accident records, cleaned 10-minute weather observations, geographic urban boundaries, annual road-section traffic statistics, and observed daily traffic counts. The primary analysis includes 5,912 rural injury accidents from 2007–2024 with valid wind observations within 20 km. Within each weather station, calendar year, and season, observed accidents in each gust interval are compared with the number expected from the local background frequency of that interval. Confidence intervals are obtained from 5,000 bootstrap samples clustered by weather station.

The observed/expected ratio is close to one across most low and moderate gust intervals. At gusts of at least 36 m/s, 25 accidents were observed compared with 4.1 expected, giving a ratio of 6.09 (95% station-clustered bootstrap interval 3.65–8.94). The result describes relative accident occurrence. It does not estimate a causal effect or accident risk per vehicle. Traffic volume is unavailable at the required spatial and temporal resolution for the full study period. A smaller 2007–2024 road-section sensitivity analysis uses estimated SDU/VDU vehicle-kilometres, while observed daily traffic counts are used separately to show that travel falls during strong winds. These checks support the relevance of traffic exposure but are too incomplete to replace the primary analysis.

The thesis documents how the data were cleaned, matched, checked, and analysed. Weather frequency, traffic exposure, coverage loss, and uncertainty are reported separately. The results are compatible with higher accident occurrence during extreme gusts. Hourly road-level traffic data would be needed for a per-vehicle interpretation.

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Chapter 1

Introduction

Strong wind can affect vehicle stability, lane keeping, driver behaviour, and the decision to travel. Iceland is a useful setting for studying this question because wind conditions vary strongly across locations and time. However, accident counts alone are not accident risk. A severe weather event can raise the risk faced by each vehicle while simultaneously reducing the number of vehicles on the road. Weather frequency and traffic exposure must therefore be treated as separate parts of the problem [Andrey et al.(2003), Lord and Mannering(2010), Washington et al.(2020)].

The original project considered many weather variables, outcomes, and model families. That scope produced numerous outputs but obscured the central question. The final study is deliberately narrower: it focuses on maximum wind gust and rural accidents involving injury. Mean wind speed and traffic data are retained as supporting analyses.

1.1 Objective and Research Questions

The objective is to determine whether rural injury accidents are over-represented during strong wind gusts after accounting for how frequently those gusts occur locally.

1. How can accident and 10-minute weather observations be linked in a reproducible documented data pipeline?
2. How does the observed frequency of rural injury accidents vary across maximum-gust intervals relative to local background gust frequency?
3. How does the interpretation change when limited traffic information is introduced as a sensitivity analysis?

The primary result estimates relative accident occurrence. It does not estimate a causal effect or a national accident rate per vehicle-kilometre.

1.2 Contribution

The work contributes a documented Python workflow and an Icelandic case study. The workflow cleans large weather files, preserves unmatched accidents, records coverage at each spatial threshold, calculates background wind frequency, produces station-clustered uncertainty intervals, and regenerates the final figures and tables. This follows the principle that computational research should be inspectable and reproducible [Peng(2011)].

Chapter 2

Data and Software Pipeline

2.1 Source Websites and Downloaded Data

The source websites and the downloaded content are listed explicitly here so that the origin of every analytical variable can be audited. Accident records were supplied as tab-separated exports from Samgöngustofa; its public accident register description and statistics page are available at `https://www.samgongustofa.is/umferd/tolfraedi/umferdarslys/` [Samgöngustofa(2026)]. Ten-minute observations and station metadata were obtained from the Icelandic Meteorological Office data service at `https://api.vedur.is/weather/` [Veðurstofa Íslands(2026)]. Annual traffic workbooks and daily counter reports were downloaded from the Vegagerðin traffic pages at `https://www.vegagerdin.is/samgongukerfid/vegakerfid/umferdartolur` and `https://umferd.vegagerdin.is/` [Vegagerðin(2026)]. Urban polygons were downloaded from the Statistics Iceland WFS endpoint `https://gis.is/geoserver/Hagstofan/ows`. Current counter locations and road-section geometry were obtained from the Vegagerðin ArcGIS layers `https://vegasja.vegagerdin.is/arcgis/rest/services/data/`.

Table 2.1: Downloaded accident, weather, and geographic source data

Downloaded content	Records	Coverage	Source columns used
Accident-event export	118,247	2007–2024	`nid`, `Dagsetning`, `Tími`, `xhnit`, `yhnit`, `meidsli`, `tegohapps`, `stadsetn`, `flokkur2`.
Vehicle export	215,982	2007–2024	`nid`, `taeki`, `eflokkur`. Used only to count vehicles in the descriptive accident profile.
Road-link export	126,610	2007–2025; analysis ends in 2024	`nid`, `dagstimi`, `borgarnumer`, `sveitavegur`. Used to identify the registered road section.
Urban-locality polygons	97 polygons	Published 2020–2024 boundaries	`stadur`, `heiti`, `lsv`, `tag`, `fjoldi_2020`, `gildirfra`, `gildirtil`, and polygon geometry.
Raw 10-minute weather	226,580,952	2007–2025	`station`, `time`, `f`, `fg`, `t`.
Weather-station metadata	771 station-history records	Station histories	`station`, `name`, `lat`, `lon`, `start`, `ending`.

Table 2.2: Downloaded traffic and road-reference source data

Downloaded content	Extracted records	Coverage	Source columns used
Annual road-section workbooks	33,757 standardised section-year records	2000–2025	Year, road number and section code, road and endpoint names, section length, `ADU`, `SDU`, `VDU`, and thousand vehicle-kilometres. Workbook column names differ by year and are standardised by the preparation program.
Daily counter PDF reports	1,033,659 channel-day values	2019–2024	Report year, `fastnr`, station identifier, road section, site name, date, and daily traffic volume. Channels at the same physical counter are subsequently summed.
Counter-location layer	1,076	Current register at download	`fastnr`, counter name, `x_3057`, `y_3057`, road number, section code, active status, and counter type.
Road-section geometry	302 study-relevant sections	Current network at download	Road and section identifiers, names, section length, `paths_json`, and geometry bounds.
Road-section midpoints	302 derived reference points	Current network at download	Road section, section length, `midpoint_x_3057`, `midpoint_y_3057`, `midpoint_lon`, `midpoint_lat`, and midpoint method.

The 2025 accident, injury, and vehicle deliveries are retained as raw source material but are not included in the 2007–2024 statistical analysis.

2.2 Accident Data

The accident source covers 2007–2024 [Samgöngustofa(2026)]. The primary study population contains 6,120 rural accidents involving injury. Accident severity is based on the source field `meidsli`: code 1 denotes a fatal accident, code 2 a serious accident, code 3 a minor-injury accident, and code 4 an accident without injury. The primary outcome is therefore `meidsli` ≤ 3. The serious-or-fatal comparison uses `meidsli` ≤ 2. The 16,487 rural damage-only accidents remain available in the source data but are not events in the injury outcome.

Rural status is calculated from accident coordinates and Statistics Iceland urban-locality polygons rather than taken directly from the original accident file [Hagstofa Íslands(2026)]. Points inside an urban polygon are classified as Urban and valid points outside all polygons are classified as Rural. A documented supervision rule classifies a defined Vestmannaeyjar area as Urban. The same 2020–2024 polygon dataset is applied to accidents from 2007–2024, so historical urban expansion is not represented. This is a limitation of the rural classification.

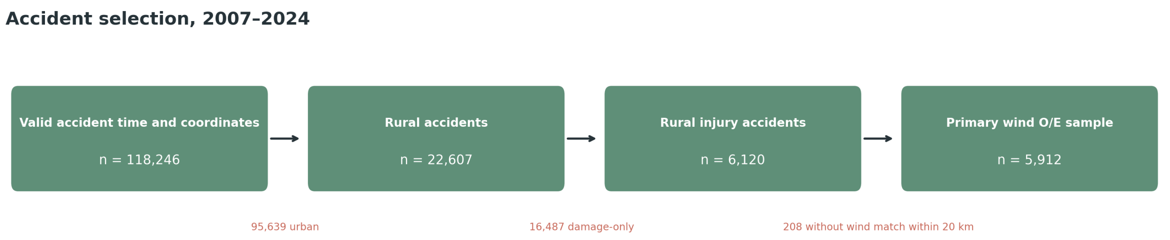


Figure 2.1: Available accident records, severity of the rural injury sample, and coverage of the primary weather match. Damage-only accidents remain available but are not part of the injury outcome.

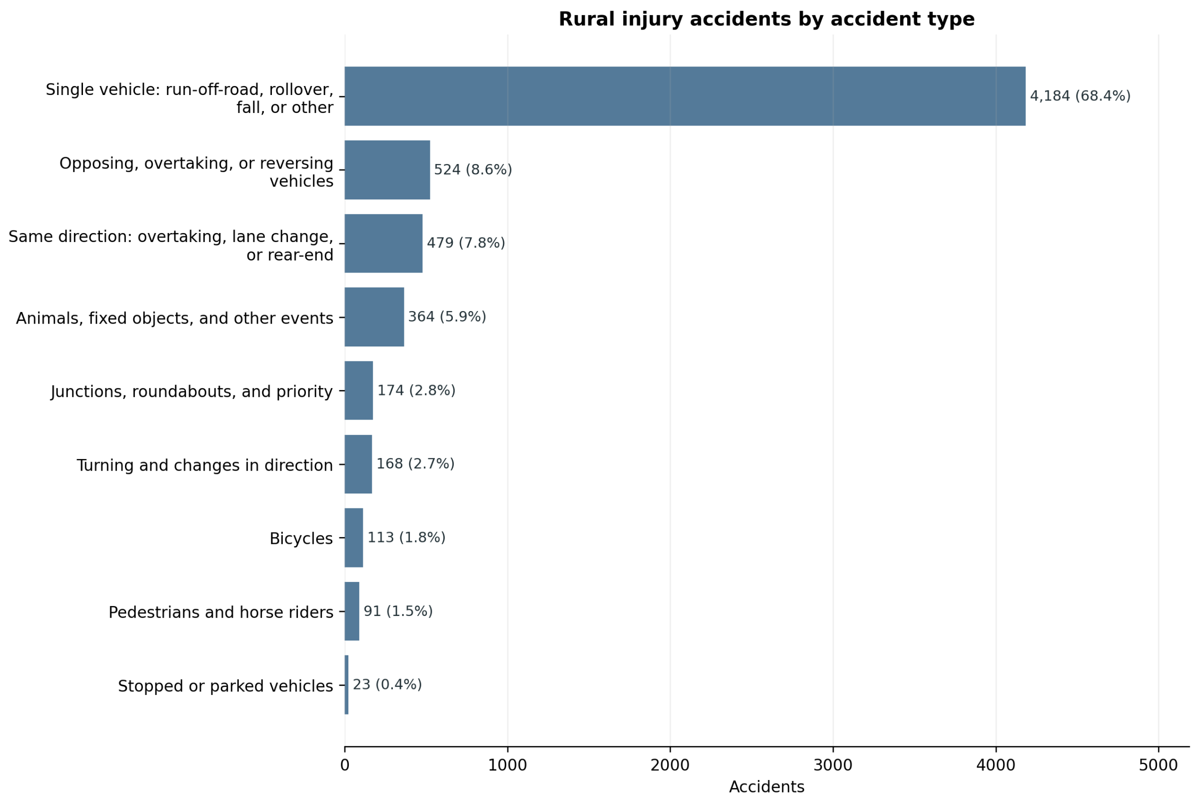


Figure 2.2: Accident-type distribution among the 6,120 rural injury accidents.

2.3 Weather Data

The weather source contains 10-minute observations from Icelandic weather stations. The main variables are mean wind speed (`f`), maximum wind gust (`fg`), and temperature (`t`). Wind rows are retained when

$$0 \leq f < 45, \qquad 0 \leq fg < 65, \qquad fg + 0.5 \geq f.$$

The raw delivery also contains station-time records from precipitation and radiation stations that do not measure wind. These station-years are reported separately rather than interpreted as failures of wind measurement. Within wind-capable station-years, rows missing either `f` or `fg` are excluded. Negative values, values at or above either upper limit, and internally inconsistent values where `fg = 0` but `f > 0` are excluded and reported in the annual audit. A value of `f = 0` alone is retained. Only uninterrupted all-zero runs where `f = fg = 0` for at least two hours are excluded as potentially frozen sensors. Temperature may be missing without invalidating an otherwise usable wind row.

Each accident is matched to the geographically nearest station with a clean wind observation at either adjacent timestamp on the 10-minute observation grid. The observations therefore have 10-minute resolution, but the selected observation is no more than five minutes from the recorded accident time. The primary maximum distance is 20 km. All 6,120 accidents remain in the output; unsuccessful matches are recorded rather than silently deleted.

Table 2.3: Weather cleaning summary, 2007–2025

Category	Records	Share of raw records
Raw station-time records	226,580,952	100.00%
Station-years without wind data	11,810,743	5.21%
Missing wind within wind-capable station-years	427,778	0.19%
Negative or upper-threshold wind values	300,649	0.13%
Frozen all-zero runs ($\geq$ 2 hours)	2,541,617	1.12%
Clean wind observations retained	211,497,897	93.34%

After station-years without wind instrumentation are set aside, 98.48% of wind-capable records are retained. The table is deliberately brief; the full audit keeps the individual quality rules separate.

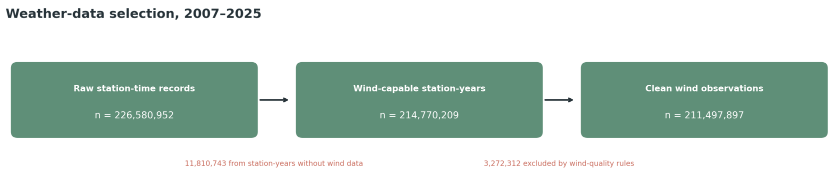


Figure 2.3: Weather observations retained after the documented wind-quality rules.

2.4 Traffic Data

Two traffic sources are used for different purposes.

Annual road-section files provide average annual daily traffic (ADU), summer daily traffic (SDU), winter daily traffic (VDU), section length, and annual vehicle-kilometres for 2000–2025 [Vegagerðin(2026)]. The traffic sensitivity uses the years shared with the accident study, 2007–2024. Following the official definitions, SDU covers June–September and VDU covers December–March. These files can be linked to accidents with a registered road-section code, but exact matching is much narrower than the weather matching.

Daily traffic counts extracted from Vegagerðin reports cover 2019–2024. They describe observed traffic at counter sites and are useful for measuring travel demand during windy days. They do not provide hourly vehicle-kilometres at every accident location. More than one counter may also occur on a road section. Each report identifies a counter by road section and *stöð* (road station). The annual road files provide the corresponding section start and end stations (`Bst` and `Est`), which permit a counter position to be interpolated along the section geometry when an official counter coordinate is not available.

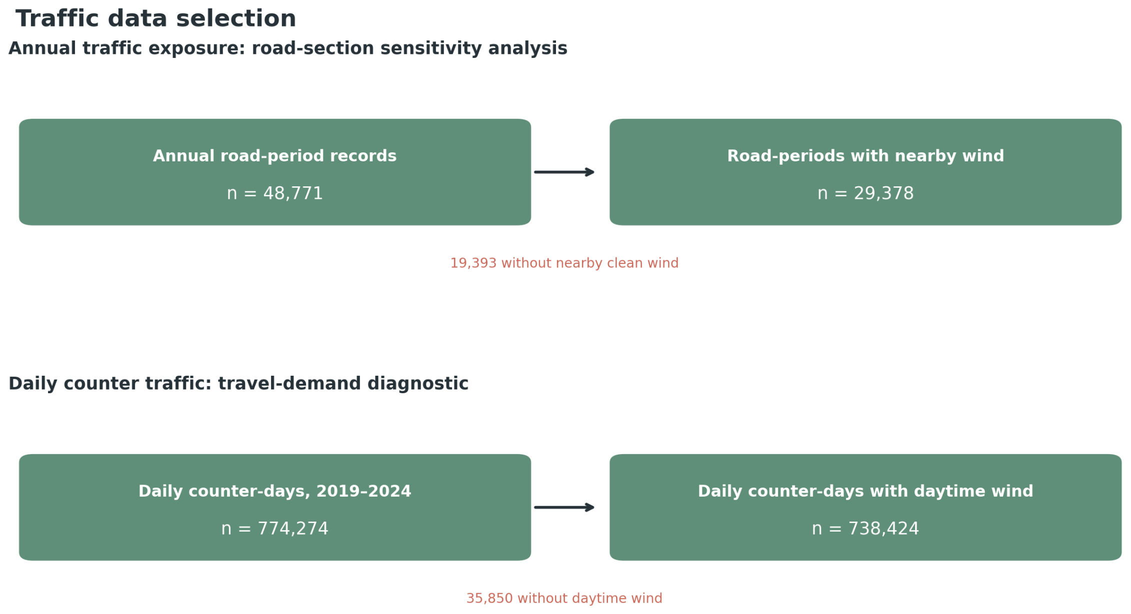


Figure 2.4: Coverage of the annual road-section data, the road-section wind table, and the separate daily-counter diagnostic.

2.5 Analysis-Ready Files

Table 2.4 lists the files created from the downloaded sources and then used directly in the statistical workflow. Record counts refer to the current reproducible build. Derived Boolean indicators and readable labels are omitted from the column lists unless they enter a reported calculation.

Table 2.4: Analysis-ready event, weather, and frequency files

File	Records	Coverage	Creation and columns used
`all_accidents_enriched.parquet`	118,247	2007–2024	Accident coordinates are transformed to EPSG:3057 and WGS84, classified against the urban polygons, and linked to the road-register export. Used: `nid`, date/time, coordinates, `meidsli`, `tegohapps`, `urban_rural`, registered road section and road number.
`rural_injury_accidents.parquet`	6,120	2007–2024	Selects Rural records with `meidsli <= 3`. Each event is assigned the nearest station having a clean adjacent 10-minute observation, subject to the reported distance limit, and its annual road section is looked up. Used: accident identifiers and time, severity and type, station identifier, distance, weather time, `f`, `fg`, `t`, road section, `ADU`, `SDU`, and `VDU`.
`weather_10min_clean.parquet`	see weather-cleaning audit	2007–2025	Created from the raw weather file by requiring non-missing wind, $0 \leq f < 45$, $0 \leq fg < 65$, and $fg + 0.5 \geq f$; negative values and zero gusts are reported in the audit. Used: `station`, `time`, `f`, `fg`, `t`.
`oe_station_period_bins.parquet`	807,743	2007–2024	Combines accident counts with background wind-bin frequency at the same station, year, and season. Used: station, year, season, variable, wind bin, measurement counts, frequency, observed accidents, expected accidents, distance rule, severity group, and time-match rule.

Table 2.5: Analysis-ready traffic and road-section files

File	Records	Coverage	Creation and columns used
annual_road_section_exposure.csv	33,757	2000–2025	Standardises all annual workbooks to one road-section-year layout. Used: year, road section, road number and names, section length, ADU, SDU, VDU, and thousand vehicle-kilometres. Statistical analyses use 2007–2024.
wind_frequency_road_period_2007_2024.parquet	220,466	2007–2024	Assigns nearby weather stations to road-section midpoints and counts valid observations by SDU/VDU period, variable, and wind bin. Used: station, year, traffic period, variable, bin limits, measurement counts, and frequency.
road_section_wind_panel_2007_2024.parquet	1,072,962	2007–2024	Joins annual traffic, road length, nearby-station wind frequency, surface lookup, and accident counts by road section, year, traffic period, and wind bin. These keys plus traffic volume, estimated vehicle-kilometres, frequency, injury counts, and serious-or-fatal counts are used.
daily_traffic.parquet	774,274	2019–2024	Sums direction/lane `fastnr` channels for the same physical counter and date. Official counter coordinates are used for conservative matches. Otherwise, the PDF *stöð* is interpolated between the same year's `Bst` and `Est` along the road-section geometry; the resulting coordinate remains explicitly marked as estimated. A section midpoint is used only when this interpolation is unavailable. Used: date, counter and road-section identifiers, traffic volume, channel count, source `fastnr`, coordinates, and location method.
daily_traffic_weather.parquet	774,274	2019–2024	Adds the nearest usable station's 10:00–21:59 mean wind and maximum gust to each counter-day. The wind-response analysis then calculates traffic relative to the mean for the same counter, year, month, and weekday. Used: counter/date, traffic, station and distance, `f_daytime_mean`, `fg_daytime_max`, typical traffic, baseline days, and traffic index.

2.6 Reproducible Implementation

The retained workflow is intentionally small. Weather cleaning is implemented in `src/clean_weather.py`; accident matching in `src/match_accidents_weather.py`; the detailed observed/expected calculation in `src/analysis/calculate_wind_risk.py`; final bootstrap tables and figures in `src/analysis/create_wind_risk_report.py`; the road-section sensitivity in `src/build_road_section_wind_table.py`; and the daily traffic diagnostic in `src/analysis/analyze_daily_traffic.py`. The retained analytical outputs can be regenerated with

```
.venv/bin/python -m src.run_analysis
```

Figures are generated directly by these programs and are not manually edited.

Chapter 3

Methods

3.1 Primary Wind-Frequency Analysis

Maximum gust is divided into 3 m/s intervals. Let g identify a weather station, calendar year, and meteorological season. Let A_{jg} be the number of accidents in gust interval j and group g, and let $A_g = \sum_j A_{jg}$. From all clean 10-minute observations, let T_{jg} be the number in interval j and T_g the total in group g. The local background frequency is

$$p_{jg} = \frac{T_{jg}}{T_g}.$$

If accident timing followed local wind frequency, the expected number in the interval would be $A_g p_{jg}$. The pooled observed/expected ratio is

$$R_j = \frac{\sum_g A_{jg}}{\sum_g A_g p_{jg}}.$$

A value of one means that the accident share equals the local time share of the gust interval. A value above one means that accidents are over-represented in that interval. Standardisation within station, year, and season prevents windy stations or seasons from dominating solely because their wind distributions differ.

Uncertainty is estimated with 5,000 weather-station-clustered bootstrap replicates. Stations are sampled with replacement, while all years, seasons, weather observations, and accidents associated with a sampled station remain together. The reported interval is the 2.5th to 97.5th percentile of the bootstrap distribution.

3.2 Why Traffic Is Not in the Primary Ratio

The primary denominator is background wind time. It does not assume that traffic is constant; traffic is simply unavailable at the required resolution. Of 5,480 accidents with an exact road-section link, 5,294 also have clean wind within 20 km. Restricting the traffic comparison to the official SDU and VDU periods leaves 3,529 accidents. April–May and October–November are excluded because their traffic would otherwise have to be derived as a residual from ADU, SDU, and VDU.

Using those traffic data in the primary curve would change the study population and mix seasonal average traffic with 10-minute accident-time gusts. The primary curve is consequently described as *wind-frequency-adjusted relative accident occurrence*. It is not a traffic-adjusted risk per vehicle.

3.3 Road-Section Traffic Sensitivity

Traffic is included where the spatial and temporal keys can be linked explicitly. For road section, year, official traffic period, and wind interval k, j, estimated vehicle-kilometres are

$$V_{kj} = q_k L_k D_k p_{kj},$$

where q_k is SDU or VDU vehicles per day, L_k is section length, D_k is the number of calendar days, and p_{kj} is the local share of clean 10-minute observations in the wind interval. The accident rate is

$$\text{Rate}_j = \frac{A_j}{\sum_k V_{kj}} \times 100{,}000{,}000.$$

For a direct comparison with the primary type of ratio, expected accidents are also allocated in proportion to V_{kj}. This method adjusts for spatial and seasonal traffic differences, but it still assumes that traffic within an SDU/VDU period is distributed across wind intervals in proportion to time. It does not observe traffic at the accident minute.

3.4 Observed Daily Traffic Diagnostic

Daily traffic is analysed separately because its weather definition is daily. The unit is one fixed counter on one date; different counters on the same road section are not summed. For each counter, year, month, and weekday, a typical traffic level is the median daily count. The traffic index is

$$I_{cd} = 100 \times \frac{Q_{cd}}{\mathrm{median}(Q_{c,\,\text{same year, month, weekday}})}.$$

An index below 100 indicates fewer counted vehicles than usual for that local calendar context. It is not a national count of people or vehicles. Daytime mean wind is calculated from clean observations between 10:00 and 21:59. Official counter coordinates are used where a conservative location link is available. Otherwise the report's *stöð* is placed proportionally between the year-specific `Bst` and `Est` on the registered road-section geometry and is marked as estimated. The nearest station within 20 km having usable observations that day is then selected. This interpretation of *stöð* is independently checked against 105 conservative name-matched official counter locations: the median position difference is 2.6 m (90th percentile 179 m). A road-section midpoint is retained only where the station range or geometry is unavailable. Uncertainty in the median index is estimated by resampling whole counters 1,000 times.

For the small residual set of midpoint-based locations, the table records the maximum possible offset if the counter were anywhere on the registered section. These are conservative location-uncertainty bounds, not estimates of the counter's actual error. Station-interpolated locations do not receive this midpoint bound: their uncertainty is historical and geometric rather than a known interval around a midpoint.

Counter-year mean traffic is compared with exact road-section ADU as a consistency check. A near-complete counter-year has at least 300 observed days; single-counter road sections are evaluated separately because multiple PDF records can represent different locations, directions, or lanes.

Chapter 4

Results

4.1 Coverage

Table 4.1: Coverage of the retained analyses

Analysis step	Available	Scope	Coverage
Wind match within 10 km	4,622	6,120	75.52%
Wind match within 20 km (primary)	5,912	6,120	96.60%
Wind match within 30 km	6,109	6,120	99.82%
Exact annual road-section match, 2007–2024	5,480	6,120	89.54%
Complete SDU/VDU traffic sensitivity	3,529	4,260	82.84%
Daily counter-days with daytime wind	738,424	774,274	95.37%

The 20 km threshold retains most accidents while limiting the spatial distance to the weather proxy. Increasing the threshold to 30 km adds 195 accidents but can be misleading in complex terrain. The traffic sensitivity is substantially narrower than the primary analysis and must not be presented as equally representative.

The two traffic rows have different analytical meanings. The first records an exact road-section and year link. The second is restricted to accidents in the eight official SDU/VDU months and also requires usable seasonal traffic and wind-frequency exposure.

4.2 Maximum Wind Gust and Relative Accident Frequency

Ratios remain close to one across most low and moderate intervals. Clear over-representation appears from 30 m/s. The strongest result is at gusts of at least 36 m/s,

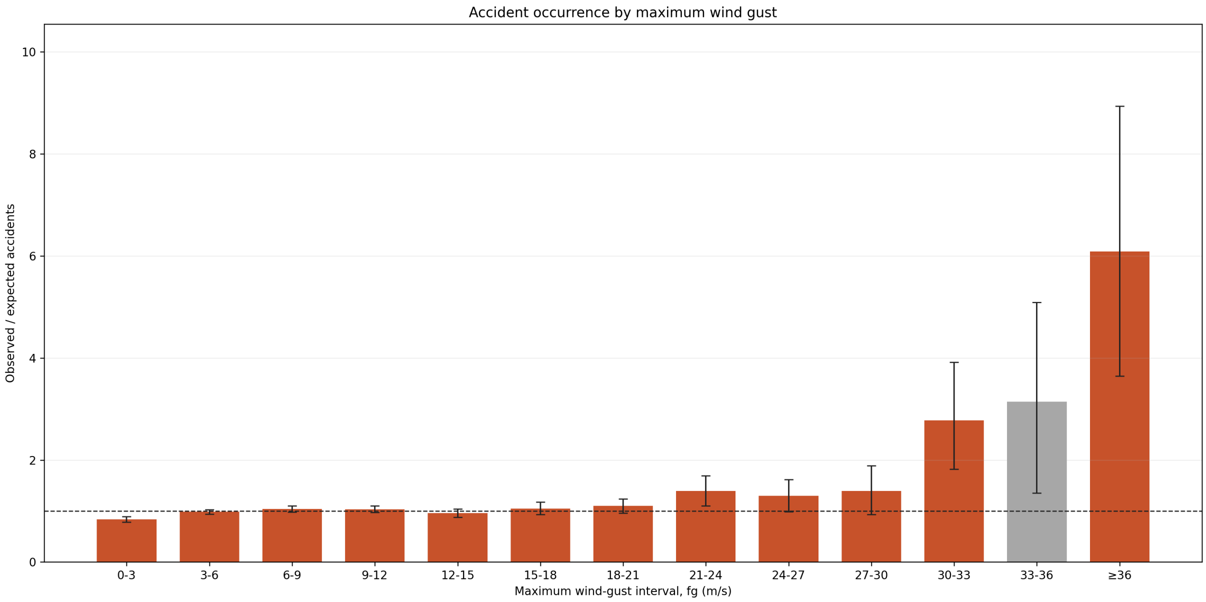


Figure 4.1: Wind-frequency-adjusted relative occurrence of rural injury accidents by maximum gust. The dashed line is the expected value of one. Confidence intervals are based on 5,000 weather-station-clustered bootstrap samples. Traffic volume is not included.

Table 4.2: Primary maximum-gust observed/expected results

Gust interval	Observed	Expected	O/E	95% clustered interval
0–3	896	1070.5	0.84	0.78–0.89
3–6	1444	1466.8	0.98	0.94–1.03
6–9	1254	1204.8	1.04	0.98–1.10
9–12	915	884.9	1.03	0.97–1.10
12–15	553	578.2	0.96	0.87–1.04
15–18	353	336.9	1.05	0.93–1.18
18–21	205	186.7	1.10	0.96–1.24
21–24	131	94.3	1.39	1.10–1.69
24–27	62	47.8	1.30	0.99–1.62
27–30	31	22.2	1.39	0.93–1.89
30–33	28	10.1	2.78	1.82–3.92
33–36	15	4.8	3.14	1.35–5.09
≥ 36	25	4.1	6.09	3.65–8.94

where 25 accidents were observed compared with 4.1 expected. The ratio is 6.09 and its station-clustered interval is 3.65–8.94. The high-wind intervals contain few accidents and should be interpreted as a general upper-tail pattern rather than as precise thresholds.

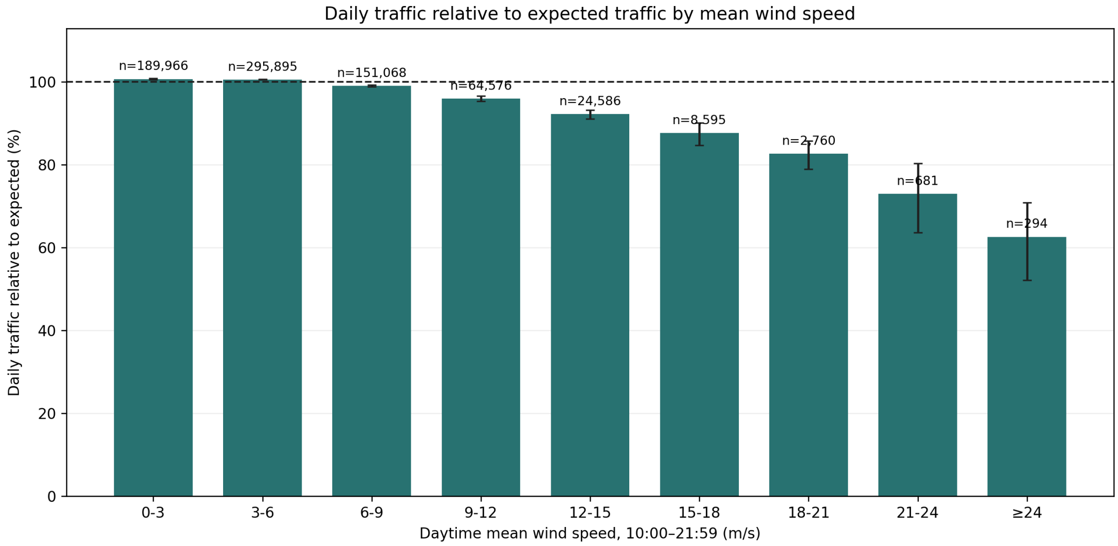
The main result is stable to station radius. For gusts of at least 36 m/s, the ratio is 7.17 at 10 km, 6.09 at 20 km, and 5.90 at 30 km. The corresponding serious-or-fatal analysis contains only four accidents in the highest gust bin; it is supportive but too sparse to replace the injury-accident result.

4.3 Mean Wind as a Supporting Analysis

Mean wind speed shows the same broad upper-tail pattern. The mean-wind O/E figure is provided in Appendix A.2; maximum gust remains the pre-specified exposure used for the main conclusion.

4.4 Daily Traffic During Wind

The PDF extraction initially contains 1,033,659 records. Directional channels at the same physical site are summed, while distinct counter stations on the same road section remain separate. This gives 774,274 physical counter-days from 476 sites. Usable daytime wind is available for 738,424 counter-days (95.37%). Traffic remains close to its local typical level below 9 m/s and declines at higher mean wind speeds. This supports lower travel demand during strong wind, but it does not estimate national traffic or traffic at each accident time. The annual-traffic sensitivity and detailed ADU validation are retained in Appendix A.3.



Expected traffic: mean for the same counter, year, month, and weekday. Daytime traffic is estimated as 95% of the 24-hour count; error bars are 95% counter-cluster bootstrap intervals.

Figure 4.2: Vehicles counted per day relative to a typical day at the same counter, year, month, and weekday. Bars are medians across fixed counters and error bars are 95% counter-cluster bootstrap intervals. Daytime mean wind is calculated from 10:00 to 21:59.

Chapter 5

Discussion

5.1 Interpretation

The primary result is clear but limited. Extreme gusts are over-represented at rural injury-accident times relative to how often those gusts occur locally. This pattern remains after the comparison is grouped by weather station, calendar year, and season, and after uncertainty is clustered by station. The result does not show that each increase in gust causes a proportional increase in accident risk. Moderate intervals fluctuate around one, and several upper intervals are sparse.

Traffic matters to the interpretation. The primary curve cannot distinguish between risk per vehicle and the number of vehicles exposed. The daily traffic diagnostic shows that travel falls in strong wind, while the small traffic-adjusted sensitivity produces larger upper-tail ratios. Together these results suggest that the primary weather-only curve is unlikely to be inflated simply because more vehicles travel during severe wind. Nevertheless, the available traffic data are too narrow to quantify the national per-vehicle effect.

5.2 Strengths

The main strengths are the large 2007–2024 accident sample, 10-minute weather resolution, explicit spatial coverage reporting, local rather than national wind-frequency standardisation, and station-clustered bootstrap intervals. The pipeline preserves unmatched records and generates its figures directly from documented tables. The reduced scope also makes the link between research question, method, and result easier to audit.

5.3 Limitations

- The primary result is not traffic-adjusted and must not be described as accident risk per vehicle-kilometre.
- A weather station up to 20 km away is a proxy for conditions at the accident location. Terrain can produce important differences over shorter distances.
- The traffic sensitivity represents 3,529 accidents in SDU/VDU months and uses estimated seasonal exposure rather than observed traffic at the accident minute.
- Daily traffic counts cover selected counter sites. Multiple counters can represent different directions, lanes, or locations; retaining counters separately avoids double counting but does not provide national coverage.
- The highest wind intervals contain relatively few accidents. Several intervals and sensitivity comparisons are examined, so isolated estimates should not be treated as independently pre-specified hypothesis tests.
- Fixed 2020–2024 urban boundaries are applied to the entire 2007–2024 accident period.
- The analysis is observational. Unmeasured road condition, vehicle type, speed, visibility, and driver behaviour can contribute to the pattern.

5.4 Next Improvements

The most valuable additional data would be verified hourly traffic counts linked to stable road-section identifiers, section geometry, and the same weather stations used in the accident analysis. That would allow accident-time gust and non-accident vehicle exposure to be measured at compatible temporal resolution. A future analysis should pre-specify wider high-wind intervals, include vehicle class and exposed-road geometry, and use a hierarchical count model once adequate traffic exposure is available. Adding more models without better exposure data would increase complexity more than credibility.

Chapter 6

Conclusion

This thesis developed a reproducible analysis of wind gusts and rural injury accidents in Iceland. Of 6,120 accidents from 2007–2024, 5,912 were matched to a valid wind observation within 20 km. After standardising for local gust frequency within station, year, and season, accidents were strongly over-represented at gusts of at least 36 m/s: 25 were observed compared with 4.1 expected, an observed/expected ratio of 6.09 with a 95% station-clustered interval of 3.65–8.94.

The finding is meaningful but not final proof of a causal per-vehicle effect. Observed daily traffic falls during strong winds, and a limited traffic-adjusted sensitivity strengthens the upper-tail pattern, but neither traffic source provides complete hourly exposure for the full study. The defensible conclusion is therefore that extreme gusts are associated with substantial over-representation of rural injury accidents relative to local wind frequency. Better linked hourly traffic data are required to convert this association into a precise road-risk estimate.

Appendix A

Supporting Analyses

A.1 Accident Characteristics

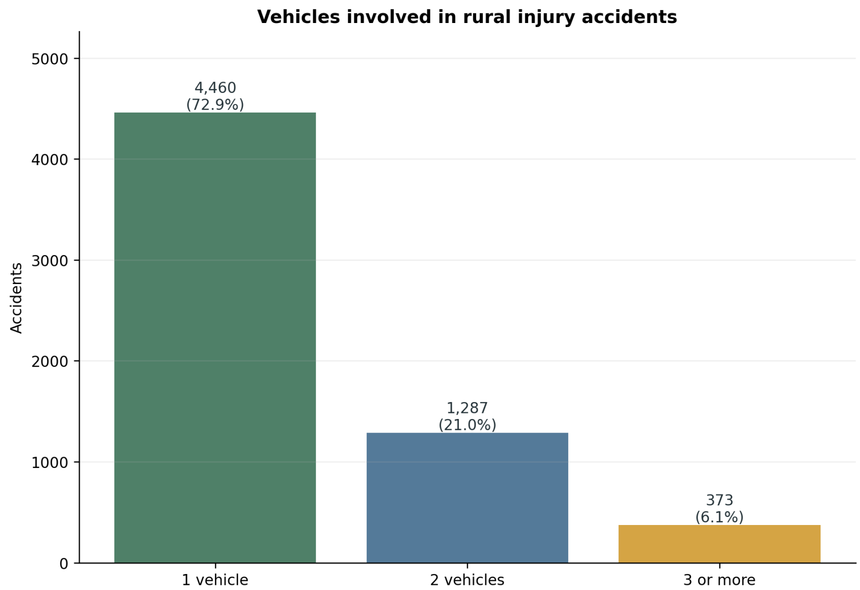


Figure A.1: Number of registered vehicles involved in each rural injury accident.

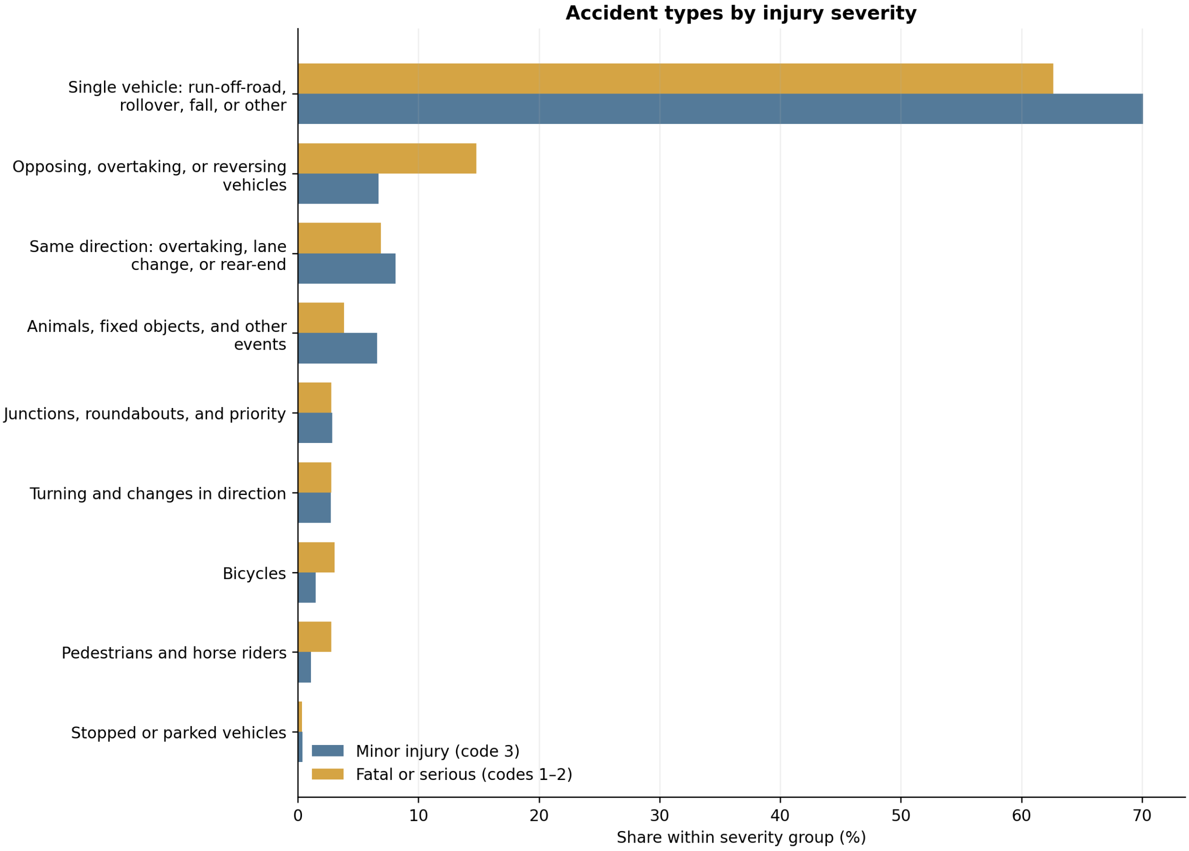


Figure A.2: Accident-type composition in minor-injury accidents and in the non-overlapping fatal-or-serious group. Percentages sum to 100 within each severity group.

A.2 Mean Wind O/E

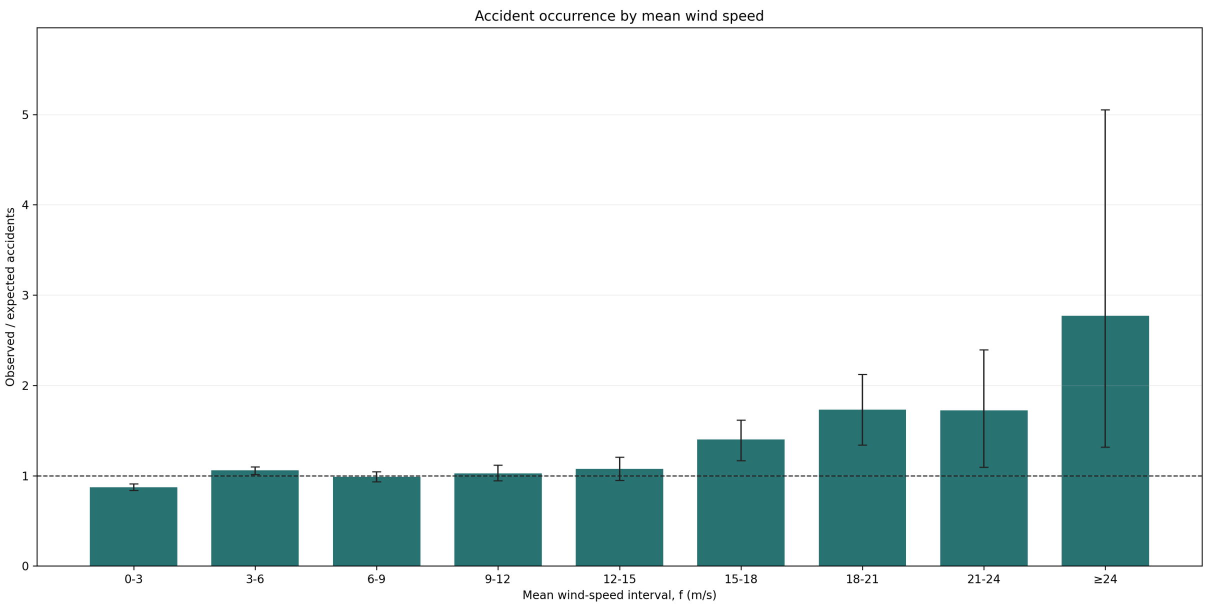


Figure A.3: Frequency-adjusted observed/expected accident occurrence by mean wind speed. This is a supporting analysis; maximum gust is the primary exposure.

A.3 Traffic Sensitivity and Validation

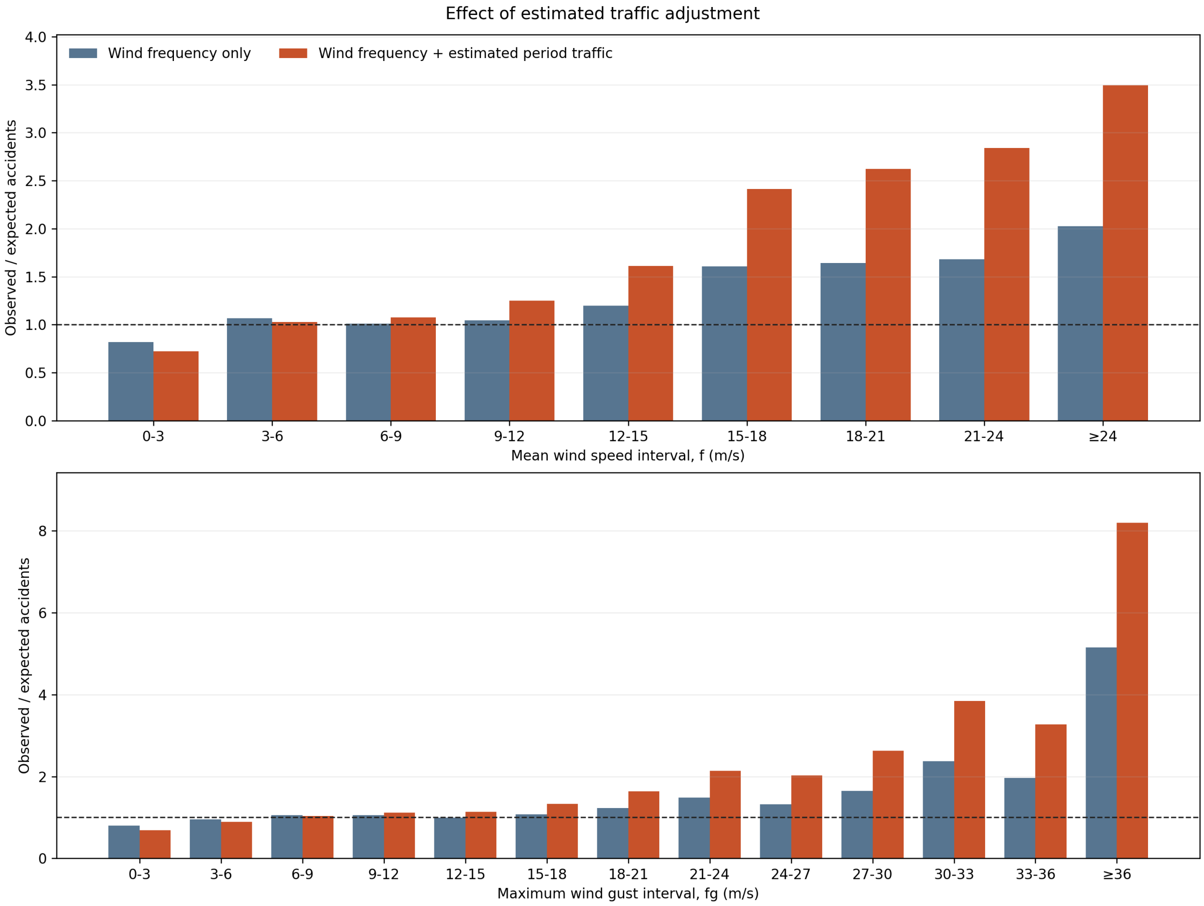


Figure A.4: Same-subset comparison of wind-frequency adjustment alone and wind-frequency adjustment plus estimated SDU/VDU traffic exposure. This restricted road-section analysis is supplementary to the primary O/E result.

Table A.1: Consistency of observed daily PDF traffic with official ADU

Scope	N	Median	P10–P90	$r_{\log}$
All exact matches	2,328	0.975	0.245–1.115	0.942
At least 300 observed days	1,906	0.984	0.240–1.063	0.940
At least 300 days, one counter per section	1,122	1.000	0.925–1.112	0.993

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